

Junqi Deng

Music AI · Audio Algorithms · Music Information Retrieval

Researcher, engineer, and musician; focused on music generation, audio intelligence, and production-grade music computing systems.

Experience

SoMuse AI | Co-founder & Chief Scientist

Jan 2026 – Present

Leading research and deployment of music-generation foundation models for the Gégé AI songwriting product.

Quwan Technology | Lyrics2Music Algorithm Lead, Changya

Feb 2024 – Apr 2025

Led technical architecture selection and model training for Lyrics2Music. The model was the first in China and the third worldwide to achieve Suno-like results; it launched in Changya's domestic product and international product, Tempolor, and won the overall championship of the Pazhou Algorithm Competition in Guangzhou.

Alibaba Group · Taobao Technology | Head of Music & Audio Effects Algorithms

Jul 2020 – Jun 2022

As a Taobao Technology Team Lead, designed and implemented the intelligent soundtrack system for Taobao short-video products, including Guangguang and Qinpai.

Developed audio-filter capabilities for Taobao Live and short-video products, including time stretch, pitch shift, voice transformation, compression, reverb, and equalization.

Developed the TaoAudio intelligent audio-interaction SDK for Taobao Live.

Alibaba Group · Innovation Business Group XLab | Algorithm Lead, JingBand Auto-accompaniment

Jul 2019 – Jul 2020

Led the design and implementation of the core JingBand auto-accompaniment engine in the JingMing app, including the accompaniment library, generation, and on-device real-time rendering.

Led the core algorithms for HeyHo AutoRap, converting ordinary speech into rap audio in the HeyHo short-video app.

Contributed to the design, implementation, and maintenance of Rhyme King, a WeChat mini-program for finding rhyming lyrics.

Alibaba Group · Digital Media & Entertainment · Xiami Music | Algorithm Lead, Music Exploration Lab

Dec 2017 – Jul 2019

Led the design and implementation of Xiami's complete audio-fingerprinting system for music recognition, blacklist monitoring, and MV-to-audio-ID matching; it remains in frequent production use within Taobao-related products.

Led Xiami Music Exploration Lab's intelligent accompaniment features, including humming and rap accompaniment, which automatically match suitable backing tracks to sung or rapped excerpts.

Designed and implemented an automatic humming accompaniment system.

Designed and implemented a word2vec-based music recommendation algorithm.

Education

The University of Hong Kong | Ph.D., Computer Engineering 2012 – 2016

Sun Yat-sen University | M.Eng., Integrated Circuit Engineering 2010 – 2012

Sun Yat-sen University | B.Eng., Microelectronics 2006 – 2010

Reviewing

IEEE Transactions on Audio, Speech and Language Processing (TASLP, 2016)

Journal of the Audio Engineering Society (JAES, 2016)

International Society for Music Information Retrieval Conference (ISMIR, 2016 – 2021)

Book Chapter & Granted Patents

[Audio Music and Computer Integration — Audio Music Technology](#), Fudan University Press, Dec 2019 (Chapters 4 and 7)

[Music Audio Generation Method and Device](#), CN112825244B, granted 2024

[Multimedia Data Generation Method, Device, Electronic Device and Computer Storage Medium](#), CN112885318B, granted 2025

Publications

Deng, J., Kwok, Y. K. [Large Vocabulary Automatic Chord Estimation Using Bidirectional Long Short-Term Memory Recurrent Neural Network with Even Chance Training](#). *Journal of New Music Research*, 47(1), 2018.

Deng, J., Kwok, Y. K. [Large Vocabulary Automatic Chord Estimation Using Deep Neural Nets: Design Framework, System Variations and Limitations](#). *arXiv:1709.07153*.

Deng, J., Kwok, Y. K. [Large Vocabulary Automatic Chord Estimation with an Even Chance Training Scheme](#). *ISMIR 2017*.

Deng, J. [Large Vocabulary Automatic Chord Estimation from Audio Using Deep Learning Approaches](#). HKU Theses Online, 2016.

Deng, J., Kwok, Y. K. [A Hybrid Gaussian-HMM-Deep Learning Approach for SeventhsBass Automatic Chord Estimation](#). *ISMIR 2016*.

Deng, J., Kwok, Y. K. [A Chord-scale Approach to Automatic Jazz Improvisation](#). Late-breaking Demo, *ISMIR 2016*.

Deng, J., Kwok, Y. K. [Automatic Chord Estimation on SeventhsBass Chord Vocabulary Using Deep Neural Network](#). *ICASSP 2016*.

Hu, X., Deng, J., Zhao, J., Hu, W., Ngai, E. C. H., Wang, R., ... & Kwok, Y. K. (2015). [SAFeDJ: A Crowd-Cloud Codesign Approach to Situation-Aware Music Delivery for Drivers](#). *ACM TOMM*, 12(1s), 21.

Hu, X., Deng, J., Hu, W., Fotopoulos, G., Ngai, E. C. H., Sheng, Z., ... & Fels, S. (2014). [SAFeDJ community: situation-aware in-car music delivery for safe driving](#). *MobiCom 2014*, 363–366.

Deng, J., Lau, F. C. M., & Kwok, Y. K. (2014). [ArmKeyBoard: A Mobile Keyboard Instrument Based on Chord-Scale System and Tonal Hierarchy](#). *ICMC 2014*.

Deng, J., Lau, F. C. M., Ng, H. C., Kwok, Y. K., Chen, H. K., & Liu, Y. H. (2014). [WIJAM: A Mobile Collaborative Improvisation Platform under Master-Players Paradigm](#). *NIME 2014*.

Earlier Projects

[Deep Learning Approaches for Large Vocabulary Automatic Chord Estimation](#) (Ph.D. thesis, 2016)

[SAfeDJ](#) (Android): an in-car music recommendation system incorporating environmental context, driver state, and music emotion ([MobiCom 2014 App Contest finalist](#))

[WIJAM](#) (iOS): a collaborative music app that uses a professional master client to give users of different skill levels a shared performance experience ([MobiCom 2013 App Contest finalist](#))

[Adding a Prefetching Pass to LLVM Polly](#): generating load-balanced, coarse-grained loop pipeline code (Google Summer of Code, 2012)

Shang: an LLVM-based high-level-synthesis compiler, focusing on automatic parallelization and data-locality optimization (M.Eng. thesis, 2012)

FPGA-based audio effect unit (reverb, delay, distortion), course project, 2011

DSP-based audio effect unit (reverb, delay, distortion), B.Eng. thesis, 2010

Personal Projects

[Web Apps](#) — A collection of web apps built with agents including OpenClaw, Hermes, Codex, and Claude Code.

[Web Media Inspector](#) — A local-first browser tool for music transcription and ear training, with audio/video loading, A/B loops, speed control, waveform navigation, EQ, and piano-roll pitch inspection.

[Lobster Podcasts](#) — Six original Chinese-language shows covering productivity, psychology, stories, biographies, interdisciplinary questions, and daily AI/market briefings; each includes RSS and a text edition.

[Open WebSSH](#) — A self-hosted, mobile-first Web SSH gateway using browser device keys; manages SSH targets, tmux sessions, and AI-agent terminals from mobile or desktop browsers.

[Web Podcasts](#) — A curated directory and player for 100 free Chinese- and English-language podcasts, with automated feed refreshes and direct playback from original publishers.

[macOS & iOS Apps](#) — Independent development of JustTimer for macOS/iOS and [JustMetronome for iOS](#). JustTimer supports menu-bar control, countdowns, breaks, preferences, and shortcuts; JustMetronome supports BPM, meter, and beat-sound configuration.

Skills

[Languages](#): C, C++, Python, Objective-C, Swift, Java, Matlab, SQL, Unix Shell

[Frameworks](#): PyTorch, TensorFlow

[Agents](#): OpenClaw, Hermes, Codex, Claude Code, OpenCode

[Music](#): Guitar, keyboards, bass, songwriting, arrangement